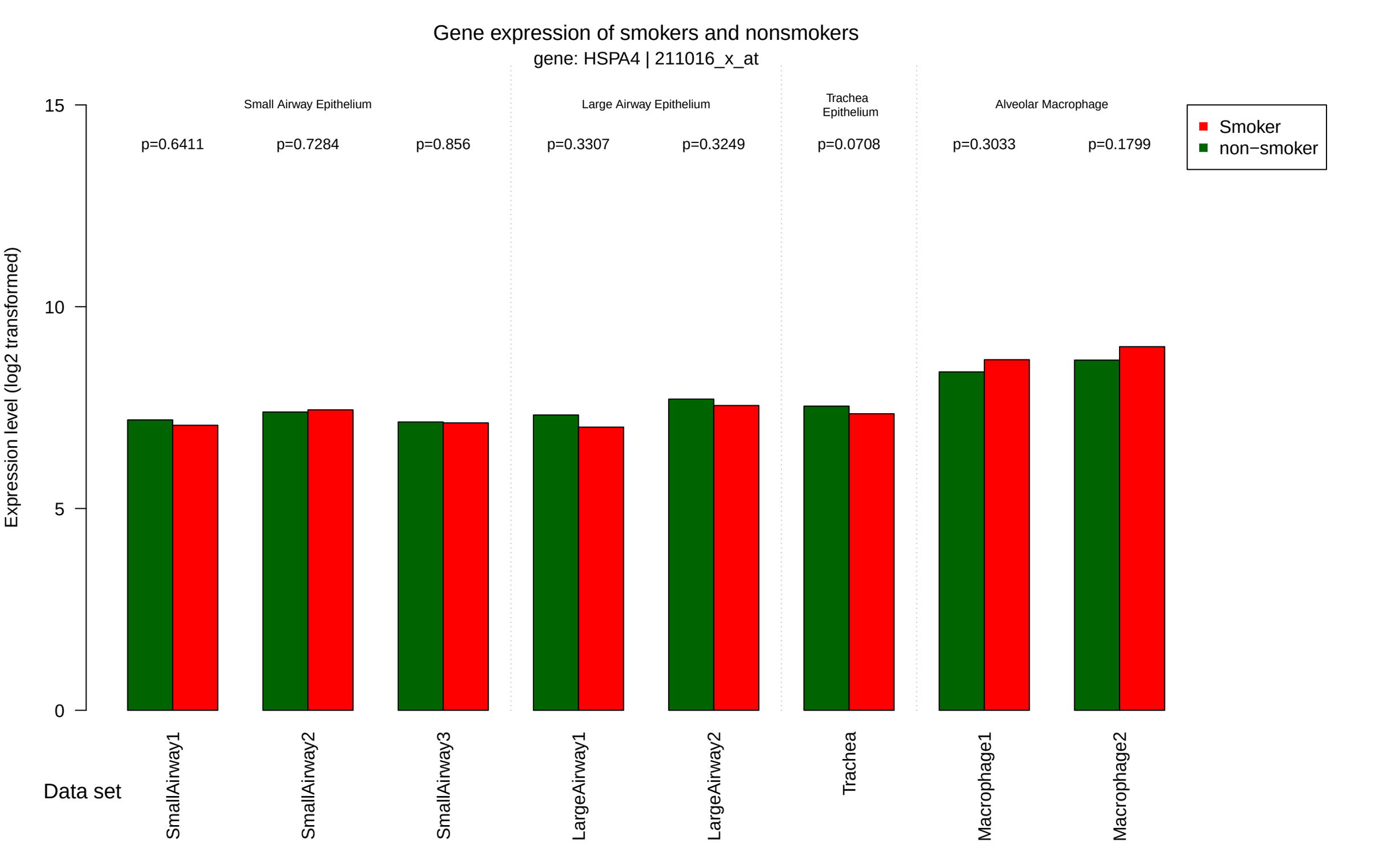
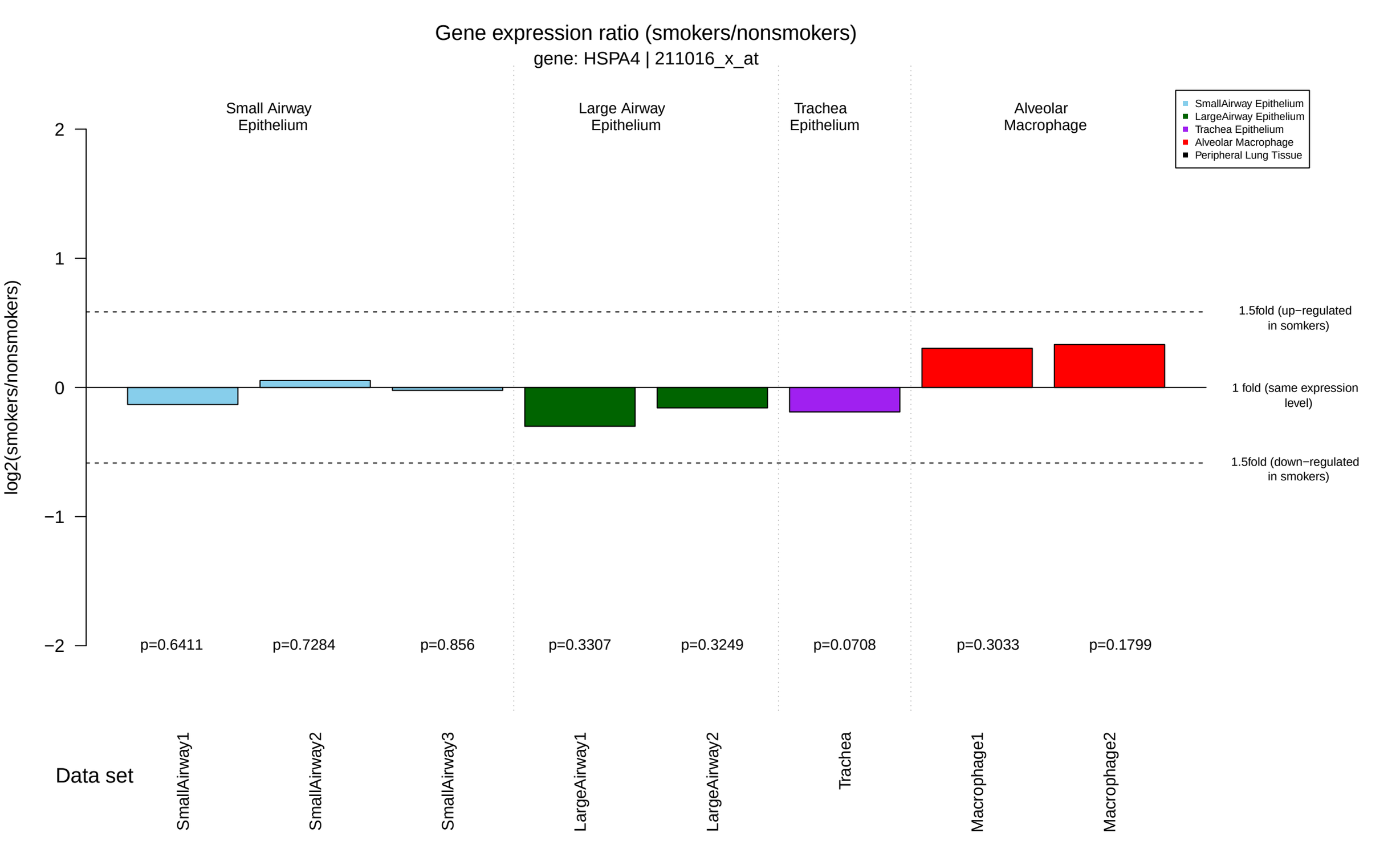
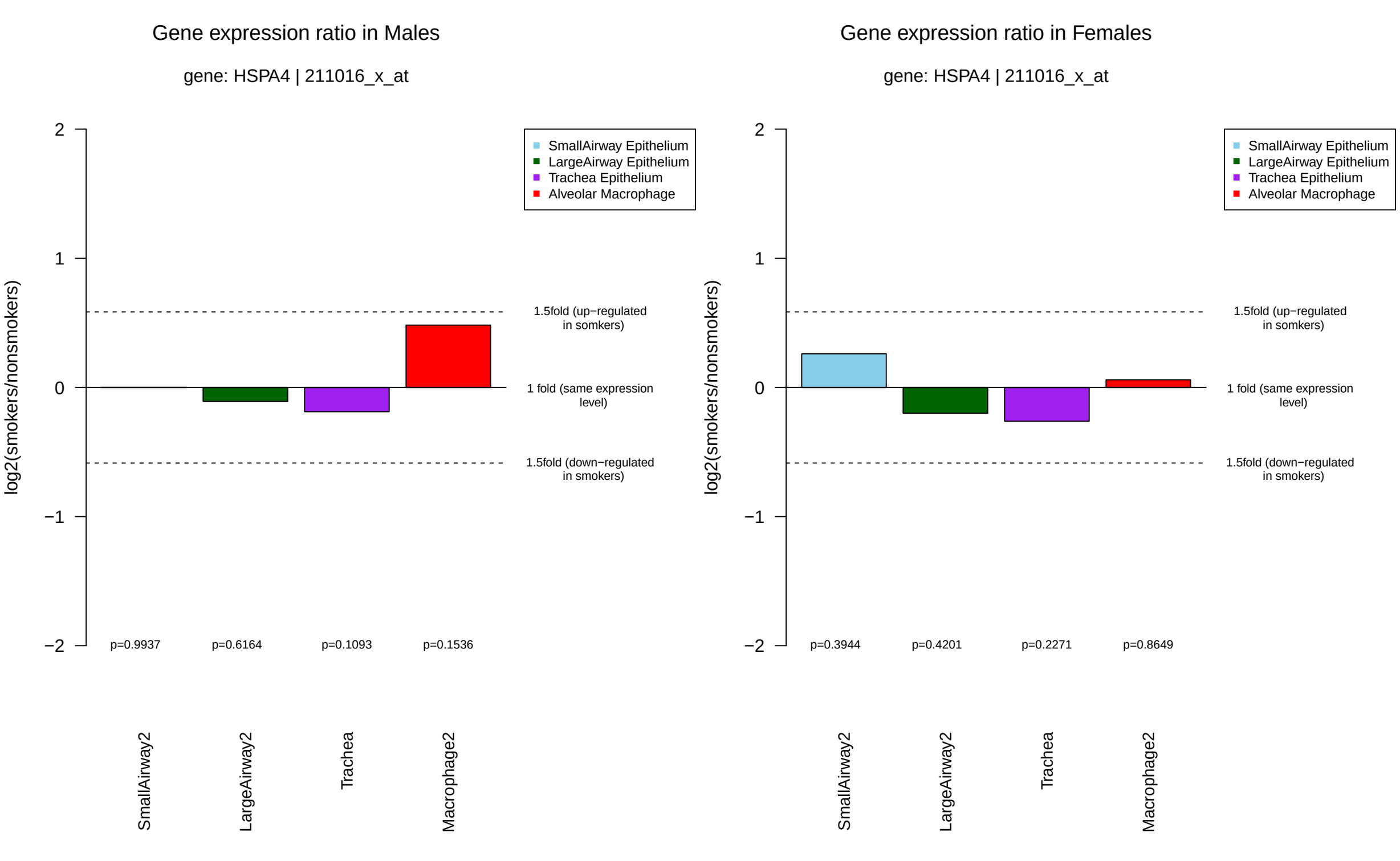
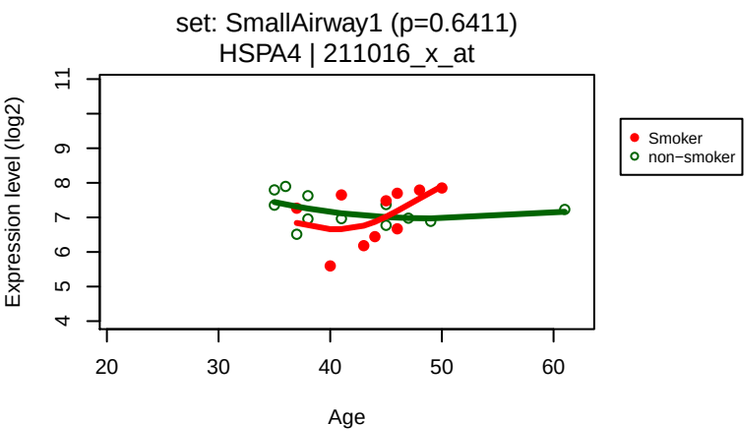
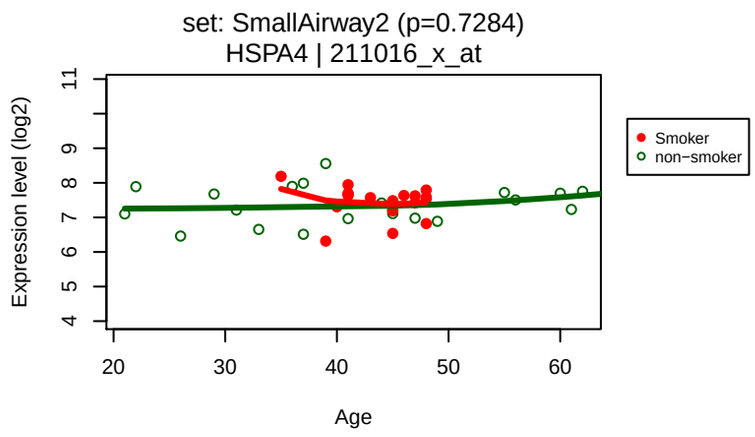
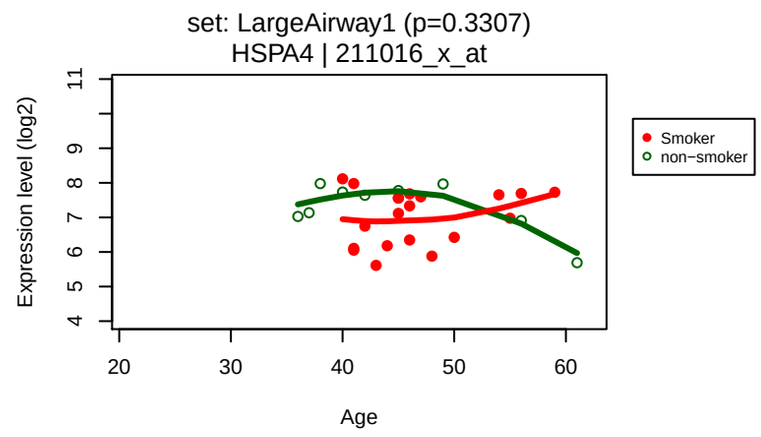

Gene expression of smokers and nonsmokers
gene: HSPA4 | 211016_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6411
p=0.7284
p=0.856
p=0.3307
p=0.3249
p=0.0708
p=0.3033
p=0.1799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

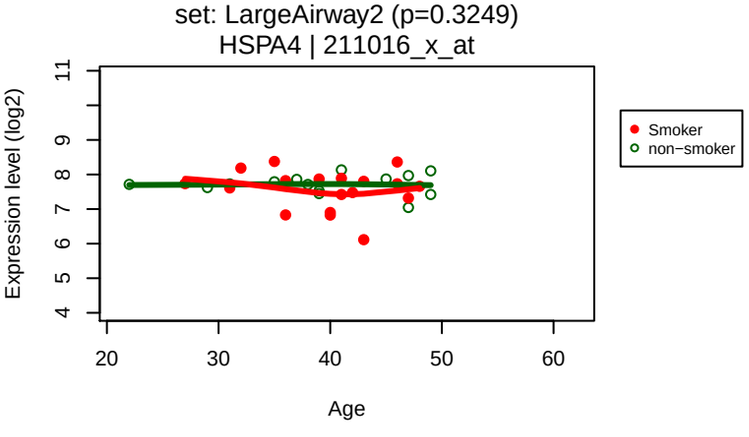

Gene expression ratio (smokers/nonsmokers)
gene: HSPA4 | 211016_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6411
p=0.7284
p=0.856
p=0.3307
p=0.3249
p=0.0708
p=0.3033
p=0.1799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

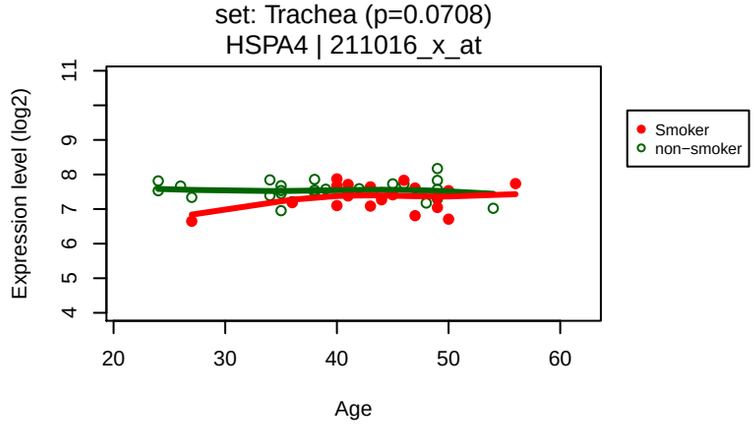

Gene expression ratio in Males
gene: HSPA4 | 211016_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9937
p=0.6164
p=0.1093
p=0.1536
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA4 | 211016_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3944
p=0.4201
p=0.2271
p=0.8649
SmallAirway2
LargeAirway2
Trachea
Macrophage2

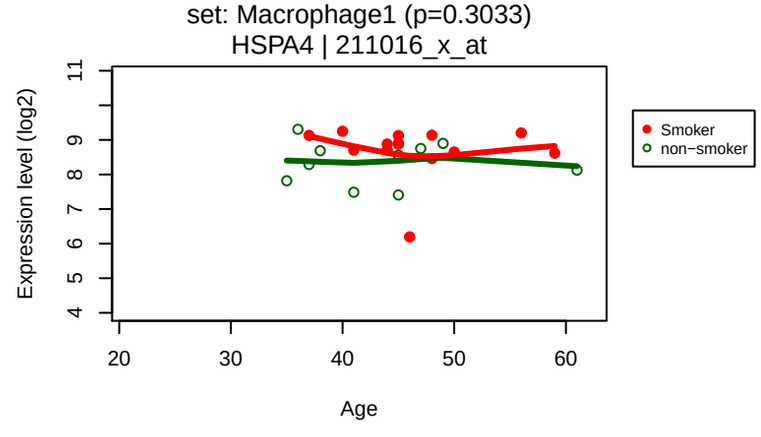

set: SmallAirway1 (p=0.6411)
HSPA4 | 211016_x_at
Expression level (log2)
Age
Smoker
non−smoker

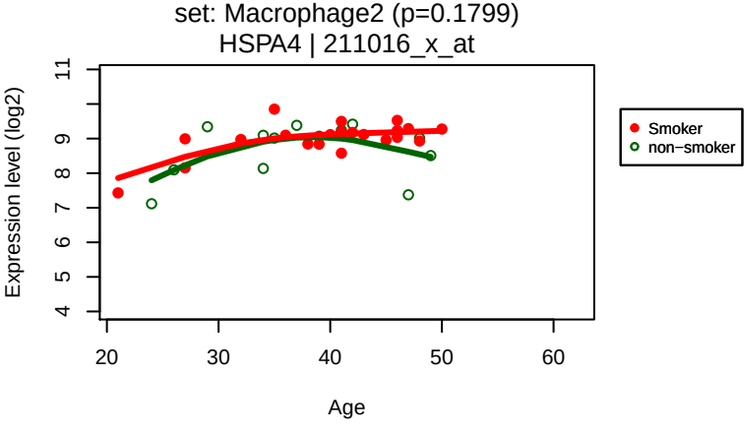

set: SmallAirway2 (p=0.7284)
HSPA4 | 211016_x_at
Expression level (log2)
Age
Smoker
non−smoker

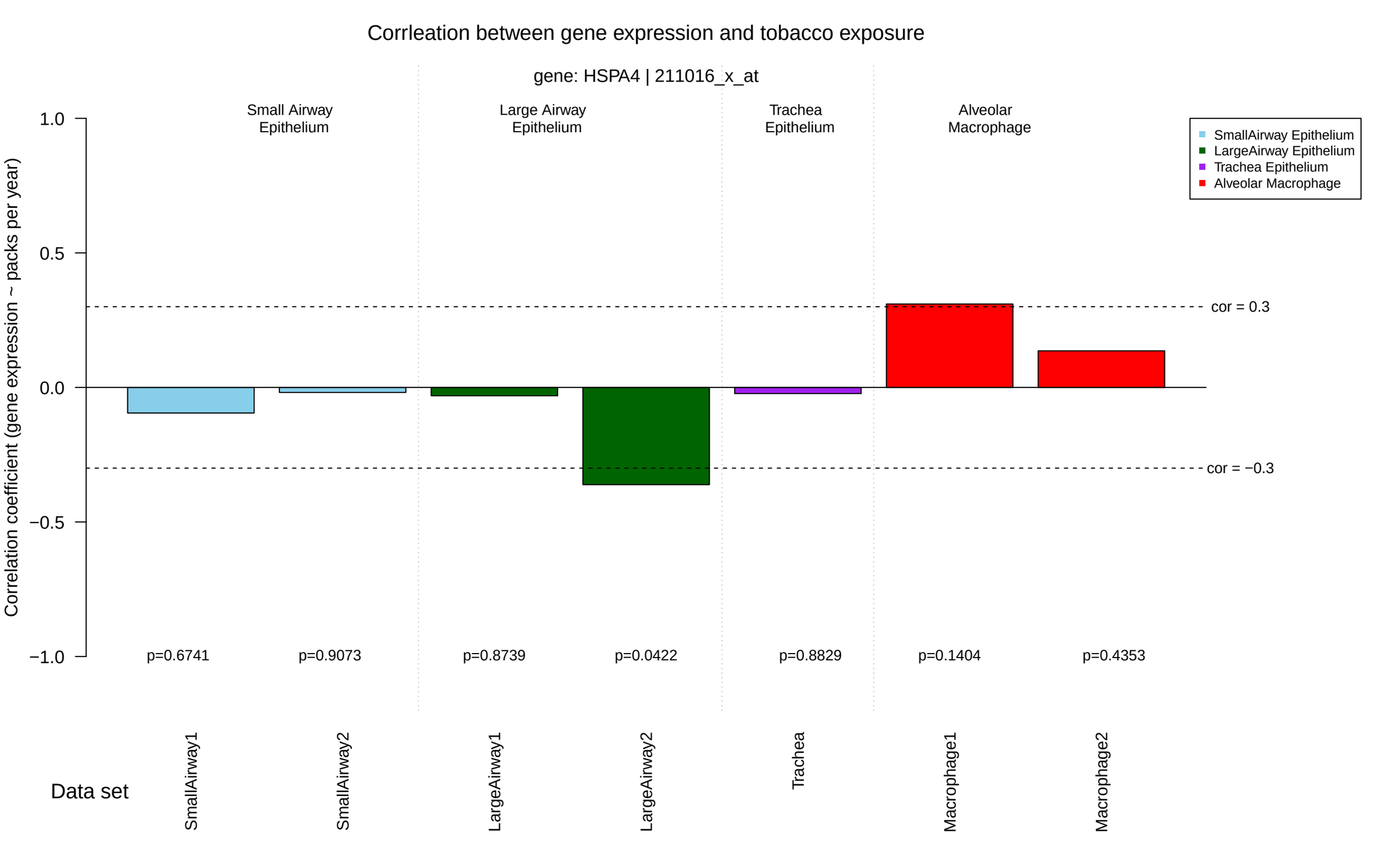

set: LargeAirway1 (p=0.3307)
HSPA4 | 211016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3249)
HSPA4 | 211016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0708)
HSPA4 | 211016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3033)
HSPA4 | 211016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1799)
HSPA4 | 211016_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA4 | 211016_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6741
p=0.9073
p=0.8739
p=0.0422
p=0.8829
p=0.1404
p=0.4353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2